

General SGD Variants

Q1: What is the fundamental difference between the parameter update of Stochastic Gradient Descent (SGD) and Mini-batch SGD?

A1: SGD updates parameters using a single data point, whereas Mini-batch SGD uses a small batch of data points.

Q2: What is the main trade-off when using SGD/Mini-batch SGD compared to Batch Gradient Descent (BGD)?

A2: SGD/Mini-batch is faster and more memory-efficient but produces noisier updates, while BGD is stable but slower.

Q3: How can the training process be improved when the gradients of the loss function are highly oscillatory or sparse?

A3: By using advanced optimizers like Momentum, RMSProp, or Adam to smooth updates.

Q4: Besides faster convergence, what is the key benefit of employing more advanced optimizers over standard SGD?

A4: They adapt the learning rate per parameter and handle sparse or complex gradients more effectively.

II. SGD with Momentum

Q5: What is the purpose of introducing a "velocity vector" in the SGD with Momentum algorithm?

A5: To accumulate past gradients and smooth parameter updates.

Q6: How does the velocity vector help in dampening oscillations during training?

A6: It reduces zig-zag movements along steep directions by combining past gradients.

Q7: How does the velocity vector aid in regions where the gradients are steady?

A7: It accelerates convergence along consistent gradient directions.

Q8: What does the momentum coefficient (ρ) control in the SGD with Momentum update rule?

A8: The weight of the previous velocity in the current update.

Q9: In terms of physical analogy, what concept does the inclusion of the velocity vector simulate in the training process?

A9: It simulates momentum in physics, like a ball rolling down a slope.

III. AdaGrad (Adaptive Gradient)

Q10: What key feature of AdaGrad distinguishes it from standard SGD with Momentum?

A10: It adapts the learning rate individually for each parameter based on past squared gradients.

Q11: How does AdaGrad adapt the learning rate for each individual parameter (β_j)?

A11: By dividing the base learning rate by the square root of the sum of all past squared gradients for that parameter.

Q12: What does $G_j^2(k)$ represent in the AdaGrad update formula?

A12: The sum of the squares of past gradients for parameter β_j up to iteration k .

Q13: In which specific scenario is AdaGrad particularly useful?

A13: In sparse data or features where some parameters require larger updates.

Q14: What is the primary drawback of the AdaGrad algorithm?

A14: The learning rate can decrease too aggressively, slowing convergence in later stages.

IV. RMSProp (Root Mean Square Propagation)

Q15: What issue of AdaGrad does RMSProp primarily address?

A15: The aggressive decay of the learning rate over time.

Q16: Instead of using the cumulative sum of past squared gradients like AdaGrad, what technique does RMSProp employ?

A16: An exponentially weighted moving average of past squared gradients.

Q17: What is the benefit of using an exponentially weighted average of past squared gradients in RMSProp?

A17: It allows the learning rate to remain adaptive without decreasing too much.

Q18: What component is typically added to the denominator of the RMSProp update rule to prevent division by zero?

A18: A small constant ϵ .

Q19: Due to the employed decay rate, why does RMSProp not suffer from the problem of reducing the learning rate too aggressively?

A19: Because the decay emphasizes recent gradients over all past gradients, preventing overly small learning rates.

V. Adam (Adaptive Moment Estimation)

Q20: What is the full name of the Adam optimization algorithm?

A20: Adaptive Moment Estimation.

Q21: Adam effectively combines the features of which two other optimization algorithms?

A21: SGD with Momentum and RMSProp.

Q22: What are the "first moment" and "second moment" that Adam tracks?

A22: The first moment is the mean of gradients (μ), and the second moment is the uncentered variance of gradients (E).

Q23: The decay rate μ is associated with the exponential moving average of which moment?

A23: The first moment (mean of gradients).

Q24: The decay rate ρ is associated with the exponential moving average of which moment?

A24: The second moment (uncentered variance of gradients).

Q25: What is the purpose of applying a "bias correction" to the estimates of the first and second moments in Adam?

A25: To counteract the initialization bias, especially in early iterations.

Q26: Write the final parameter update equation for $\beta_j^{(k+1)}$ in the Adam algorithm.

A26: $\beta_j^{(k+1)} = \beta_j^{(k)} - \eta * (\hat{\mu}_j^{(k)} / (\sqrt{\hat{E}_j^{(k)} + \epsilon}))$

VI. Other Optimization Techniques

Q27: The algorithms like L-BFGS include information on derivatives of what order?

A27: Second-order derivatives (Hessian information).

Q28: What is the key advantage of using optimization algorithms that include the Hessian?

A28: They converge faster by considering curvature information.

Q29: What is the primary disadvantage of using Hessian-based optimization compared to first-order methods like Adam?

A29: High computational cost and memory requirements for large models.

Q30: In the context of model training, what is the term for the collection of data points used to compute the gradient in Mini-batch SGD?

A30: Mini-batch.